

USPTO Patent Claims: Topological Modeling and Crossing Number Optimization

Inventors: Xavier Callens, Socrate AI

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Background

The present invention relates to advanced systems for computational optimization in graph theory and algebraic geometry, with applications in quantum circuit layout and cryptographic system modeling.

Claims

1. A computer-implemented method for optimizing a physical quantum circuit layout, the method comprising:

- (a) Receiving a digital representation of qubit interactions;
- (b) Constructing a complete graph model K_n wherein said qubits are represented as vertices;
- (c) Executing, via a hardware processor executing a Rust-based mathematical kernel, a crossing number approximation algorithm utilizing Zarankiewicz conjecture bounds to generate a constrained topological embedding;
- (d) Generating a physical layout configuration that minimizes SWAP gate insertions based on the embedded crossing number.

2. A system for accelerated modeling of topological systems for cryptography, comprising:

- A data store containing geometric definitions of Calabi-Yau five-fold manifolds;
- A symbolic and numerical computation engine configured to apply c_5 period identities to reduce integral complexity;
- A high-performance numerical compute cluster executing said reduced-complexity expressions to generate properties of post-quantum cryptographic lattices.